



RetailStack POS - 10 Week Roadmap (Full Custom API Build)

This roadmap outlines how to build a white-label POS with custom API based on Loyverse-style features, with backend modules and endpoints distributed over 10 weeks.

Week 1 – Project Setup & Auth/Tenant System

- - Set up Turborepo monorepo (apps/api, apps/web, packages/shared).
- - Integrate node-auth-prisma-boilerplate for authentication.
- - Implement multi-tenant system: tenant creation, JWT tenant scoping.
- - Endpoints: POST /auth/register, POST /auth/login, POST /tenants, GET /tenants/:id

Week 2 – Product Catalog & Categories

- - Design product schema and relationships (name, SKU, price, tax, tenant ID).
- - Implement CRUD for products and categories.
- - Endpoints: GET/POST/PUT/DELETE /products, GET/POST /categories

Week 3 – Inventory Management

- - Implement stock tracking per product.
- - Create inventory logs for tracking changes (manual or via sale).
- - Endpoints: PATCH /products/:id/stock, GET /inventory/logs

Week 4 – Sales System & Receipts (POS Flow)

- - Build sales recording system with transaction logging.
- - Generate digital receipts and sales breakdown.
- - Endpoints: POST /sales, GET /sales/:id, GET /receipts

Week 5 – Customer Management

- - Add customer profiles with contact info, total spend, visit count.
- - Endpoints: GET/POST/PUT/DELETE /customers

Week 6 – Employee Roles & Access Control

- - Set up employee onboarding and roles (e.g., Cashier, Admin).
- - RBAC with middleware for endpoint protection.
- - Endpoints: POST /employees, GET /employees, PATCH /employees/:id/role, GET/POST /roles

Week 7 – Basic Reporting System

- - Design and generate core reports: daily sales, inventory status.
- - Endpoints: GET /reports/sales, GET /reports/summary, GET /reports/inventory

Week 8 – Offline Mode Foundations

- - Integrate IndexedDB (on frontend) to store product & sales data.
- - Build background sync logic for when connection is restored.

Week 9 – Receipt Printing & Device Integration

- - Support thermal receipt printing (browser-based).
- - Test with USB/Bluetooth printers.

Week 10 – QA, Polish, MVP Demo

- - Run integration & user tests (Cypress or Playwright).
- - Polish UI/UX, add branding support per tenant.
- - Deploy backend (Render/Railway) and frontend (Vercel/Netlify).
- - Prepare MVP demo + feedback review.